

Lecture 14: Concentration Inequalities, Weak Law of Large Numbers

Jamie Haddock

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Concentration inequalities::: {.callout-warning icon=false}## Theorem: Markov's InequalityFor $X \geq 0$, $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

... . ::: {.callout-warning icon=false}## Theorem: Chebyshev's InequalityApplying Markov to $(X - \mu)^2$: for $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

...—::: {.callout-warning icon=false}## Theorem: Weak Law of Large NumbersIf $X_1, X_2, \dots$ iid with mean μ , variance σ^2 , and $\bar{X}_n = \frac{1}{n} \sum X_i$, then for every $\epsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \rightarrow 0.$$

...:

Proof:

$\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$, so Chebyshev gives $P(|\bar{X}_n - \mu| \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \rightarrow 0$.

0.1 Chebyshev's bound vs. reality

```
mu, sigma2 = 5, 9n = 30means = np.array([rng.normal(mu, np.sqrt(sigma2), size=n).mean() for _ in range(n)])
```

—# Case Study: A Concentration Guarantee for Trace Estimation $\text{Var}(\bar{T}) = \text{Var}(T)/m$ combined with Chebyshev gives a sample-size guarantee: to ensure $P(|\bar{T} - \text{tr}(A)| \geq \epsilon) \leq \delta$, take

$$m \geq \frac{\text{Var}(T)}{\epsilon^2 \delta}.$$

```
n = 60A = rng.normal(size=(n, n))true_trace = np.trace(A)z_single = rng.choice([-1, 1], size=(4000, n))
```

Enjoy Fall Break! When we return: convergence in probability and the CLT.## Reading & HomeworkReading: §5.1-5.2 (Bertsekas & Tsitsiklis, *Introduction to Probability*).Due today: Homework 6 (Gradescope).